

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Robotics Software Engineer

Robotics Software Engineer specialising in perception, SLAM, and real-time deployment of computer vision systems on constrained edge hardware. Experienced in developing ROS2-based vision pipelines and deploying quantised CNN models across safety-critical and harsh field environments. Skilled in integrating LiDAR-camera data for localisation and mapping, with proven success delivering production-level AI modules for industrial and field-tested systems. Proficient in ROS2, OpenCV, PCL, Docker, and OpenVINO, with hands-on experience in Gazebo simulation, SLAM optimisation, and edge deployment for robust, autonomous operation under demanding real-world conditions.

WORK EXPERIENCE

Esbaaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Engineered production-grade AI software for real-time perception and safety analytics, delivering deterministic performance across 24/7 oil rig operations.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Integrated AI modules with production systems (Docker, FastAPI, Kafka, Git CI/CD) in collaboration with software/hardware/QA teams to deliver fault-tolerant deployments.
- Deployed and validated the full system in harsh field environments, ensuring software robustness, traceability, and repeatable performance under variable conditions.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Built and optimised deep learning pipelines with YOLOv5 and DeepSORT, enabling consistent object tracking and robust analytics in dynamic industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing throughput by 233% and enabling real-time detection of safety-critical events.
- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Implemented rigorous testing and validation frameworks simulating challenging lighting, occlusion, and motion conditions, improving CNN stability and reducing variance by 25%.
- Mentored engineers on optimisation and deployment practices, raising code quality and delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and RGB data for autonomous navigation on an Unmanned Ground Vehicle (UGV).
- Processed and analysed large-scale point cloud datasets using Open3D and RTAB-Map to perform 3D feature extraction, registration, and map optimization.
- Tuned SLAM parameters to improve localisation accuracy across varied terrains, achieving stable performance in both feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS 2 before successful real-world deployment.
- Integrated multiple sensing modalities to achieve robust spatial and radiological understanding, enabling accurate obstacle detection in static environments.
- Implemented waypoint navigation in simulation using ROS2 Nav2 (standard planners/controllers) within Gazebo, validating end-to-end navigation workflows.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Perception & Computer Vision:** 3D Feature Extraction, Data Augmentation, Denoising, LiDAR SLAM, NumPy, Open3D, OpenCV, Point Cloud Library, Registration, Scikit-Image, Vision Transformer (ViT)
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), FastAPI, Git, Intel VPUs, Linux, Microsoft Azure, MLFlow, NVIDIA Jetson
- **Programming & Development:** Bash, C++, Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Python, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Core Machine Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks (YOLOv5/v8, EfficientDet), DeepSORT, MultiRocket, ONNX, OpenVINO, Optuna, TensorRT